山东建筑大学　　计算机学院　　实验报告

班级：　软件 212　姓名：　孙帅康　　学号：202111110337　　实验成绩：

课程：　软件项目管理　　同组者：尹皓琪，王潇文，邵亚琦，姜魁哲

实验日期：2024.3.23

实验三 服务调用

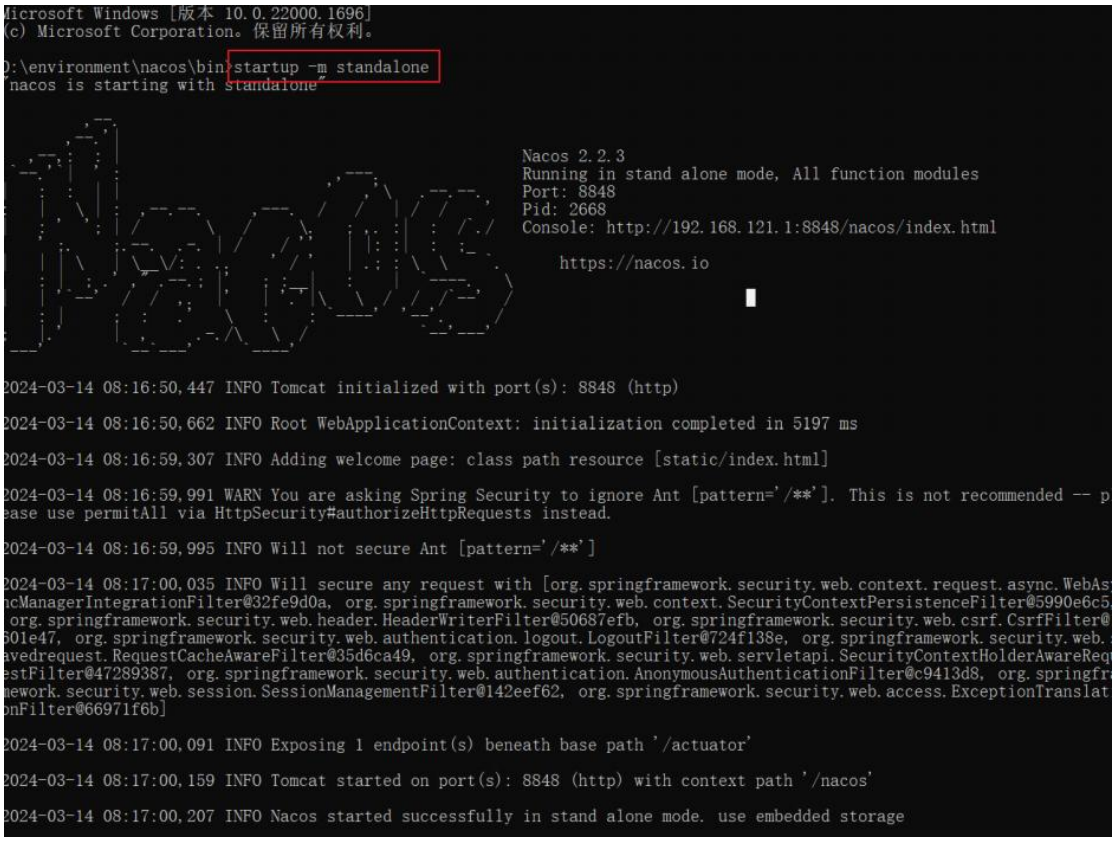
一、实验目的

掌握远程服务调用的方法。

二、实验内容

通过 restTemplate 实现远程调用。

1. 启动 nacos



2. 建立项目，对 pom.xml 文件进行配置

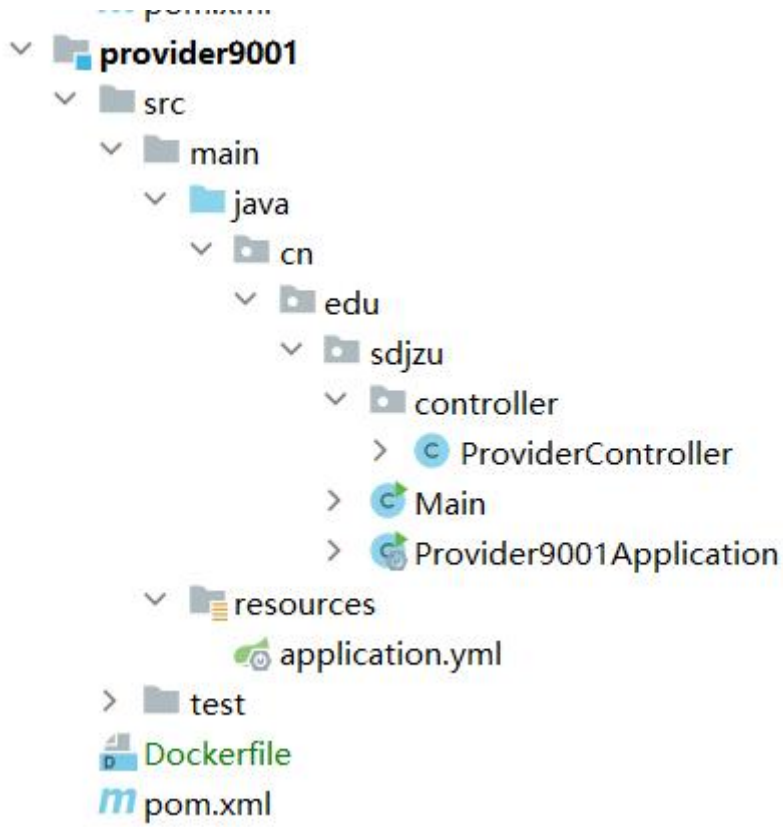
添加依赖

```
<dependencyManagement>
    <dependencies>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-dependencies</artifactId>
            <version>3.0.2</version>
```

```
            <type>pom</type>
            <scope>import</scope>
        </dependency>

        <dependency>
            <groupId>org.springframework.cloud</groupId>
            <artifactId>spring-cloud-dependencies</artifactId>
            <version>2022.0.0</version>
            <type>pom</type>
            <scope>import</scope>
        </dependency>

        <dependency>
            <groupId>com.alibaba.cloud</groupId>

<artifactId>spring-cloud-alibaba-dependencies</artifactId>
            <version>2022.0.0.0</version>
            <type>pom</type>
            <scope>import</scope>
        </dependency>

        <dependency>
            <groupId>com.mysql</groupId>
            <artifactId>mysql-connector-j</artifactId>
            <version>8.3.0</version>
        </dependency>

        <dependency>
            <groupId>org.mybatis.spring.boot</groupId>
            <artifactId>mybatis-spring-boot-starter</artifactId>
            <version>3.0.3</version>
        </dependency>

        <dependency>
            <groupId>org.projectlombok</groupId>
            <artifactId>lombok</artifactId>
            <version>1.18.24</version>
        </dependency>
    </dependencies>
</dependencyManagement>
```

```
<build>
    <plugins>
        <plugin>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-maven-plugin</artifactId>
            <configuration>
                <excludes>
                    <exclude>
                        <groupId>org.projectlombok</groupId>
                        <artifactId>lombok</artifactId>
                    </exclude>
                </excludes>
            </configuration>
        </plugin>
    </plugins>
</build>
```

3.在项目下建立模块 provider9001



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4.对 provider9001 的 pom 文件进行配置

添加依赖：

```
 <dependencies>
       <dependency>
           <groupId>com.alibaba.cloud</groupId>

<artifactId>spring-cloud-starter-alibaba-nacos-discovery</artifac
tId>

       </dependency>
       <dependency>
           <groupId>org.springframework.boot</groupId>
           <artifactId>spring-boot-starter-actuator</artifactId>

       </dependency>
       <dependency>
           <groupId>org.springframework.boot</groupId>
           <artifactId>spring-boot-starter-web</artifactId>
           <version>3.0.2</version>
       </dependency>
    </dependencies>
```

5.在 resources 文件夹下建立 application.yml 文件

```
server:
  port: 9001
spring:
  application:
    name: provider9001
  cloud:
    nacos:
      discovery:
        server-add: 192.168.23.1:8848
        service: provider
management:
  endpoints:
    web:
      exposure:
        include: '*'
```

6.建立主程序 Provider9001Application.java 文件

```
package cn.edu.sdjzu;
import
org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.boot.builder.SpringApplicationBuilder;
@SpringBootApplication
public class Provider9001Application {
    public static void main(String[] args){
        new
SpringApplicationBuilder(Provider9001Application.class).run(args);
    }

}
```

7.新建包controller在包下建立ProviderController.java文件

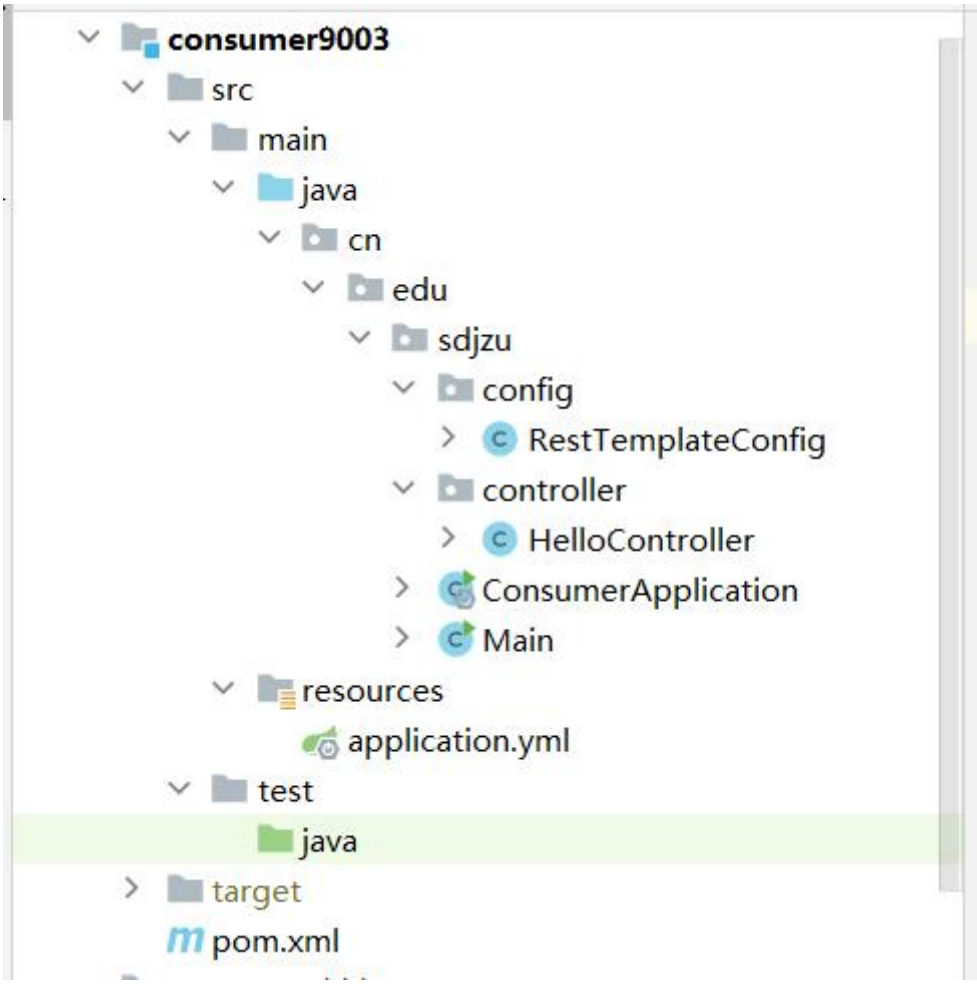
```
package cn.edu.sdjzu.controller;



import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;
import org.springframework.web.bind.annotation.RequestMapping;
import org.springframework.web.bind.annotation.RequestMethod;
@RestController
public class ProviderController {
     @RequestMapping(value="/hello",method = RequestMethod.GET)
    public String hello(){
        return "hello world,provider9001";
    }
}
```

8.用同样的方法步骤建立provider9002模块

9.建立模块consumer9003



10.对 pom.xml 文件进行配置

```
  <dependencies>
        <dependency>
            <groupId>com.alibaba.cloud</groupId>
            <artifactId>spring-cloud-starter-alibaba-nacos-discov
ery</artifactId>

        </dependency>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-actuator</artifactId>

        </dependency>
        <dependency>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-starter-web</artifactId>
```

```
            <version>3.0.2</version>
        </dependency>
        <dependency>
            <groupId>org.springframework.cloud</groupId>
            <artifactId>spring-cloud-starter-loadbalancer</artifa
ctId>
        </dependency>
    </dependencies>
```

11.在 resources 文件夹下建立 application.yml 文件

```
server:
  port: 9003
spring:
  application:
    name: consumer9003
  cloud:
    nacos:
      discovery:
        server-add: 192.168.23.1:8848
        service: consumer
management:
  endpoints:
    web:
      exposure:
        include: '*'
```

12.建立主程序 ConsumerApplication.java 文件

```
package cn.edu.sdjzu;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplicati
on;
import org.springframework.cloud.client.discovery.EnableDiscovery
Client;

@SpringBootApplication
//@EnableDiscoveryClient
public class ConsumerApplication {
    public static void  main(String[] args){
        SpringApplication.run(ConsumerApplication.class,args);
```

```
    }
}
```

13.建立包 config 在包下建立 RestTemplateConfig.java 文件

```
package cn.edu.sdjzu.config;

import org.springframework.cloud.client.loadbalancer.LoadBalanced;
import org.springframework.context.annotation.Bean;
import org.springframework.context.annotation.Configuration;
import org.springframework.web.client.RestTemplate;

@Configuration
public class RestTemplateConfig {
    @Bean
    @LoadBalanced
    public RestTemplate restTemplate(){
        return new RestTemplate();
    }
}
```

14.建立包 controller 在包下建立 HelloController.java 文件

```
package cn.edu.sdjzu.controller;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.GetMapping;
import org.springframework.web.bind.annotation.RestController;
import org.springframework.web.client.RestTemplate;

@RestController
public class HelloController {
    @Autowired
    private RestTemplate restTemplate;
    @GetMapping("/hello")
    public String sayHello(){
        String url = "http://provider/hello";
```

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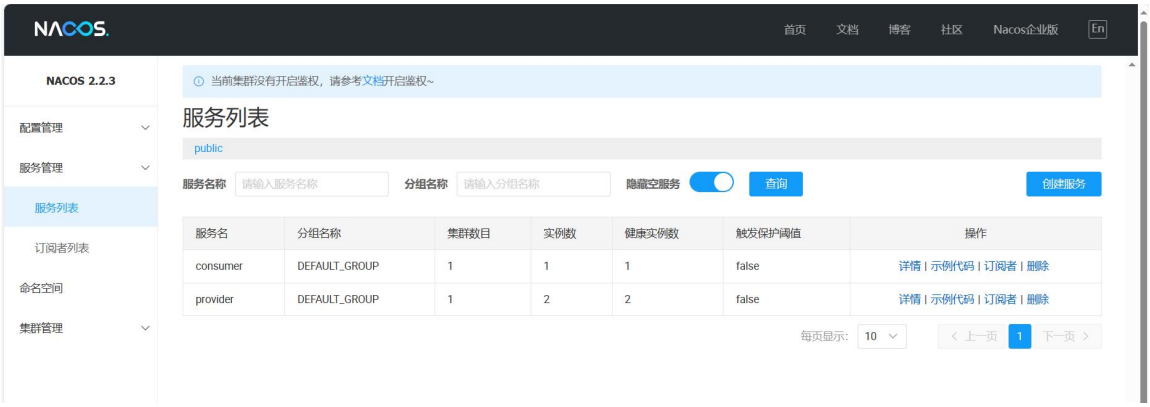
```
        String result =restTemplate.getForObject(url,String.class)
;
        result = result +"  It is from consumer application!";
        return result;
    }

}
```

15. 浏览器效果显示：

点击刷新会两个程序交错运行

nacos 显示



三、实验总结

通过本次实验学会了服务调用，负载均衡。学会了通过 restTemplate 实现远程调用。